1.

(1)miss penalty

(2)hit time

(3)miss rate

(4)hit time

(5)miss rate

(6)miss penalty

(7)miss rate

2.

(a) 1word = 4byte -> byte offset= 2 bits

(b)$\frac{128\text{KB}}{16\text{B}} = \frac{128 \times 1024}{16} = 2^{13}$ -> set field=13bits

(c)4 word per block -> word offset = 2bits

Tag field = 32-13-2-2=15bits

3.

Block number $= \frac{256}{16} = 16$

Set number $= \frac{16}{2} = 8$

Byte address	Block address	Tag	Index	H/M	Write back
0	00000	00	000	Miss	No
1	00000	00	000	Hit	No
256	10000	10	000	Miss	No
128	01000	01	000	Miss	Yes
0	00000	00	000	Miss	No

4.

(1)Block number = word address % 16

16 one word block -> cache index = 4 bits

Word address	Block number	Tag	Index	H/M
00010	2	0	0010	M
00011	3	0	0011	M
00100	4	0	0100	M
10000	0	1	0000	M
10010	2	1	0010	M
10000	0	1	0000	H
00100	4	0	0100	H
00010	2	0	0010	M

(2)Block number = word address % 4

4 words in one block -> offset=2bits

4 four-word block -> cache index=2bits

Word address	Block number	Tag	Index	H/M
00010	2	0	00	M
00011	3	0	00	H
00100	0	0	01	M
10000	0	1	00	M
10010	2	1	00	H
10000	0	1	00	H
00100	0	0	01	H
00010	2	0	00	M

(3)Block number = word address % 16

Four way set -> cache index = 2 bits

Word address	Block number	Tag	Index	H/M
00010	2	000	10	M
00011	3	000	11	M
00100	4	001	00	M
10000	0	100	00	M
10010	2	100	10	M
10000	0	100	00	H
00100	4	001	00	H
00010	2	000	10	H